as late side effects of RS. There were no severe side effects of MS. The survival starting at RS was 10.9+ 9.2 (mean) months with a median survival of 9 months (1.2–44 months). Twenty seven of 35 patients (37%) died: 8 patients (22.8%) died of progressive brain metastases, 1 died of meningeal carcinomatosis (2.9%), 15 patients (42.6%) died of extra-cerebral progressive disease and in 3 cases (8.6%) the cause of death is unknown.

Conclusions: This is the first study demonstrating the equivalence of RS and MS in the same group of patients. Tumor control and treatment tolerance after RS was comparable with tumor control and treatment tolerance after MS.

2120 Preference Elicitation Using the Time Trade-Off Method in Patients with 1 to 3 Newly Diagnosed Brain Metastases in a Prospective Randomized Clinical Trial Comparing Radiosurgery With and Without Whole Brain Radiation

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Purpose/Objective: Preference elicitation is used to address the value of healthcare outcomes. Patient preferences for health states were compared in brain metastasis patients randomized to be treated with radiosurgery (SRS) with or without whole brain radiation (WBRT). The objectives of the preference elicitation study were to 1) evaluate differences in utilities between treatment groups and 2) to evaluate changes in utilities over time.

Materials/Methods: A time trade-off instrument was developed to directly elicit preferences from patients on the trial. The elicitation instrument included a written script, data collection form, and visual aid to facilitate the understanding of trading time. The instrument was administered face-to-face during patient clinic visits at baseline, 1, 2, 4, 6 months, and every 3 months thereafter. Patients traded time based on their preference for a fixed amount of time in their current state of health (reference period) versus a reduced amount of time in optimal health. The elicitation procedure was framed using three different reference periods - 10-year, 5-year, and 1-year. Mean utilities were evaluated between treatment groups (SRS vs. SRS + WBRT), demographic variables, and within subjects over time.

Results: The mean age of patients in the trial was 61 years (range = 32–80). Twenty-four patients received SRS and 17 received SRS and WBRT. The majority of patients had lung cancer as their primary tumor (61%) and most had only one metastatic lesion to the brain (68.3%). Of the 43 patients enrolled in the trial, 41 participated in the elicitation exercise. The mean baseline utility over all reference periods was 0.87 and the mean 6-month follow-up utility over all reference periods was 0.82. There were no significant differences in mean utilities from baseline to six-month follow-up between treatment groups or demographic variables. There were significant differences in mean utilities for all patients across different reference periods. Utilities were significantly higher when using the1yr reference period (1yr reference period mean = 0.91, 5yr reference period mean = 0.88, 10yr reference period mean = 0.84; adj. $F = 4.2$, $p = 0.049$). The same effect was observed for one-month follow-up utilities (1yr reference period mean = 0.97, 5yr reference period mean = 0.87, 10yr reference period mean = 0.85; adj. $F = 5.3$, $p = 0.03$). At the one-month follow-up, mean utilities were significantly higher for those receiving SRS without WBRT using the 1yr reference period (SRS group mean = 1.0; SRS+WBRT group mean = 0.93; $F = 4.7$, $p = 0.042$). At the two-month follow-up, mean utilities were significantly higher for patients having their primary tumor site in the lung versus all other primary tumor sites (lung primary mean utilities = 0.93, other primary mean utilities = 0.71; $F = 5.0$, $p = 0.037$).

Conclusions: Utilities for patients in this trial were relatively stable and remained high suggesting reasonable preference for their current health state at the time of elicitation. Most patients did not trade time in their current health state for less time in a more optimal health state when the trade-off was presented with a short 1-year reference period. This utility analysis will provide data needed to perform cost-utility analysis of SRS vs. SRS and WBRT in the future.

REFERENCE PERIOD	BASELINE	1 MONTH	2 MONTH	4 MONTH	6 MONTH
10 year	0.83 (n=38)	0.85 (n=24)	0.83 (n=21)	0.81 (n=15)	0.81 (n=9)
5 year	0.88 (n=39)	0.87 (n=24)	0.86 (n=21)	0.84 (n=15)	0.82 (n=9)
1 year	0.91 (n=38)	0.97 (n=24)	0.87 (n=21)	0.87 (n=15)	0.84 (n=9)

2121 Choroidal Melanoma Treated With Iodine-125 Episcleral Plaque: Analysis of Dose on Disease Control and Visual Outcome

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Purpose/Objective: The Collaborative Ocular Melanoma Study (COMS) group randomized trial has established the use of episcleral I125 plaques as an alternative treatment to enucleation for medium sized choroidal malignant melanoma (CMM). The COMS protocol utilizes a dose of 85 Gy to the tumor apex or a minimum 5 mm height, which ever is greater. The optimal prescribed dose to ensure acceptable disease free survival, yet minimize radiation related ocular complications, is uncertain. We have often used doses that are lower than the COMS group trial and have prescribed dose to the tumor apex even when tumors were < 5 mm in height. We herein report our experience, specifically assessing the impact of dose on outcome.

Materials/Methods: All patients treated at our institution for CMM between 1988 through 2001 with I125 plaques were retrospectively reviewed. Preoperative and postoperative tumor characteristics, treatment related side effects, and best-corrected visual acuity were assessed by regular follow-up. Poor visual outcome was defined as a visual acuity decrease of ≤4 lines on